WVC 2026 · XXI WORKSHOP ON COMPUTER VISION

Maritime Search and Rescue

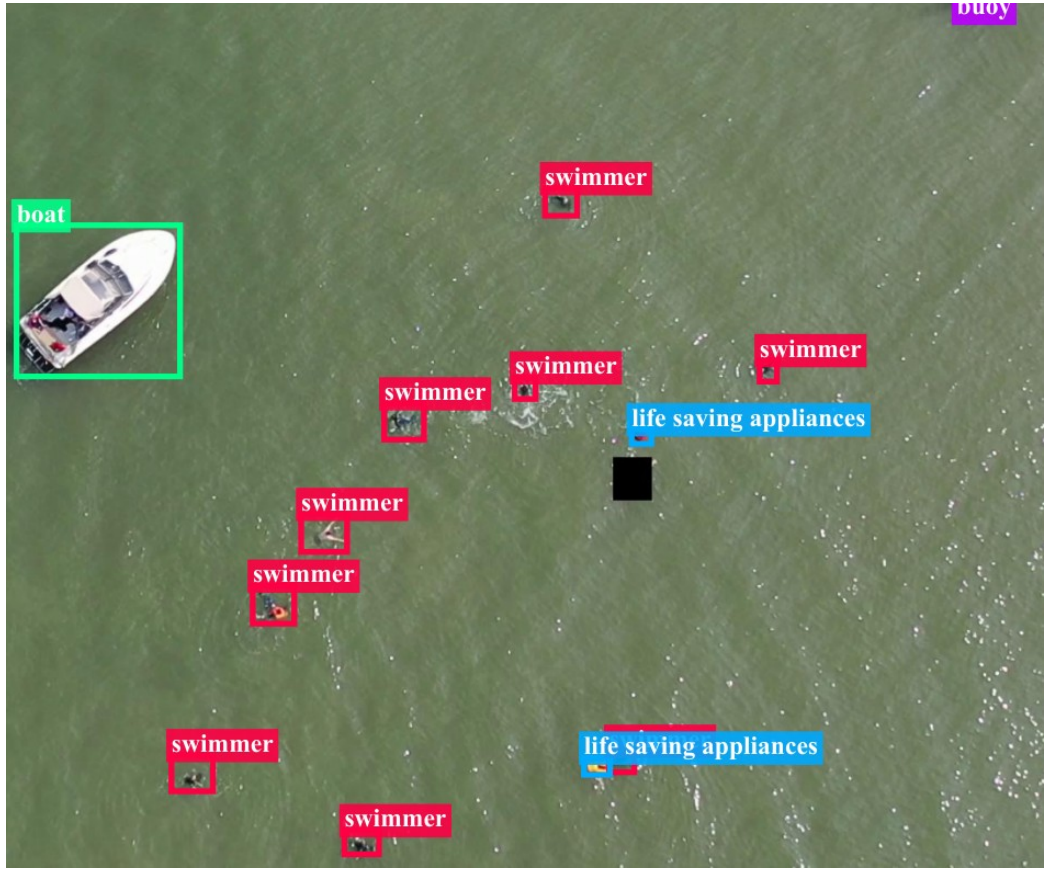
An Evaluation of Small-Object Detection in UAV Imagery

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follow the slides

THE PROBLEM: TARGETS OF A FEW PIXELS



SeaDronesSee Object Detection v2, ground-truth labels (Fig. 1 of the paper)

The same target, in fewer pixels



Original crop | Target at ~16 px | Target at ~9 px

Illustration: the same crop, resampled.

16 px
median target size at 1280 px

9.4 px
life-saving appliances

Half of the targets fit inside a 16 × 16-pixel square

SEADRONESSEE: IMBALANCED CLASSES

8,930
training images

1,547
validation images

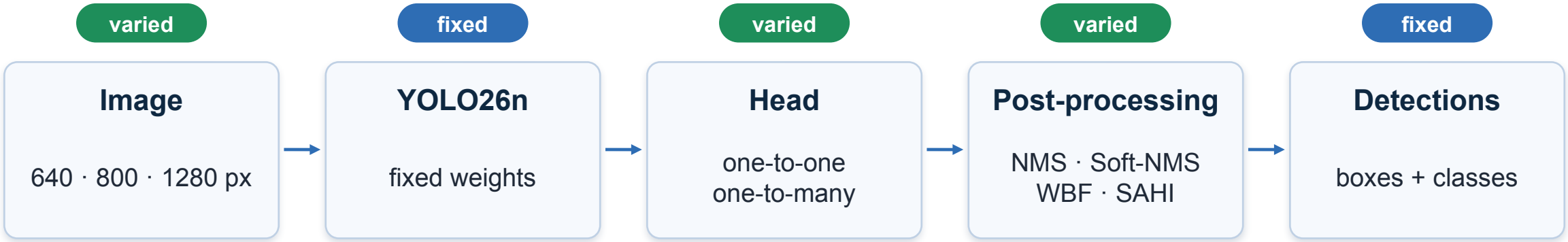
3,750
test images (official server)

Share of the training instances



The rarest class is exactly the most critical one in SAR

WHAT CHANGES: INFERENCE ONLY



How we measure

COCO AP
local val and official test server

ms/image
latency and forward passes per image

95% CI
paired bootstrap and operating points

Same architecture: which inference gives the best accuracy–cost trade-off?

TWO HEADS, THE SAME WEIGHTS

One-to-one (default)

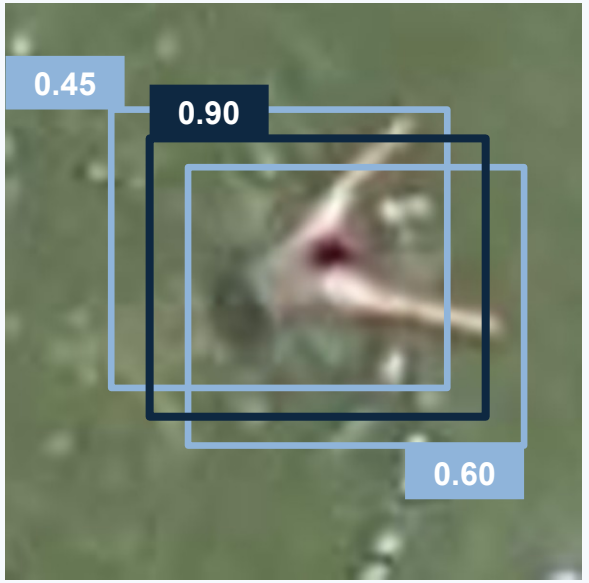


1 prediction per object

no NMS

direct output

One-to-many



several candidates per object

needs NMS

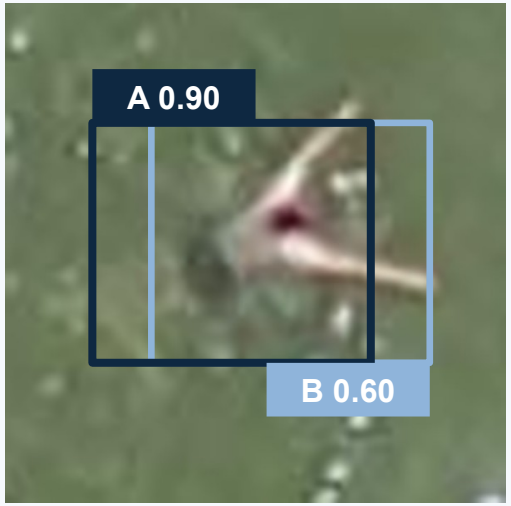
candidates → NMS → output

Illustrative boxes and confidences.

Same training, two inference paths: without NMS or with NMS

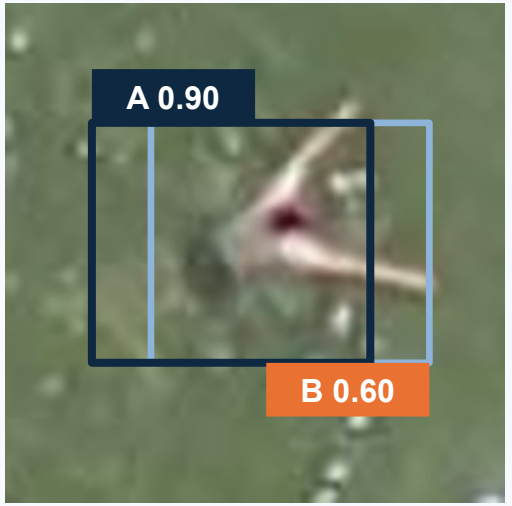
HARD-NMS VS. SOFT-NMS

Candidates



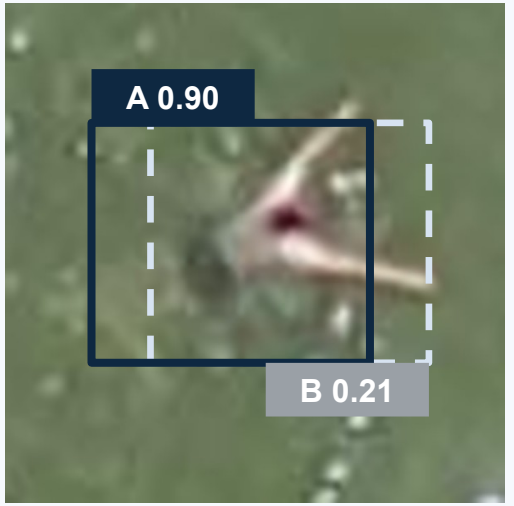
IoU(A, B) = 0.65

Hard-NMS (threshold 0.80)



0.65 < 0.80: B stays intact

Linear Soft-NMS

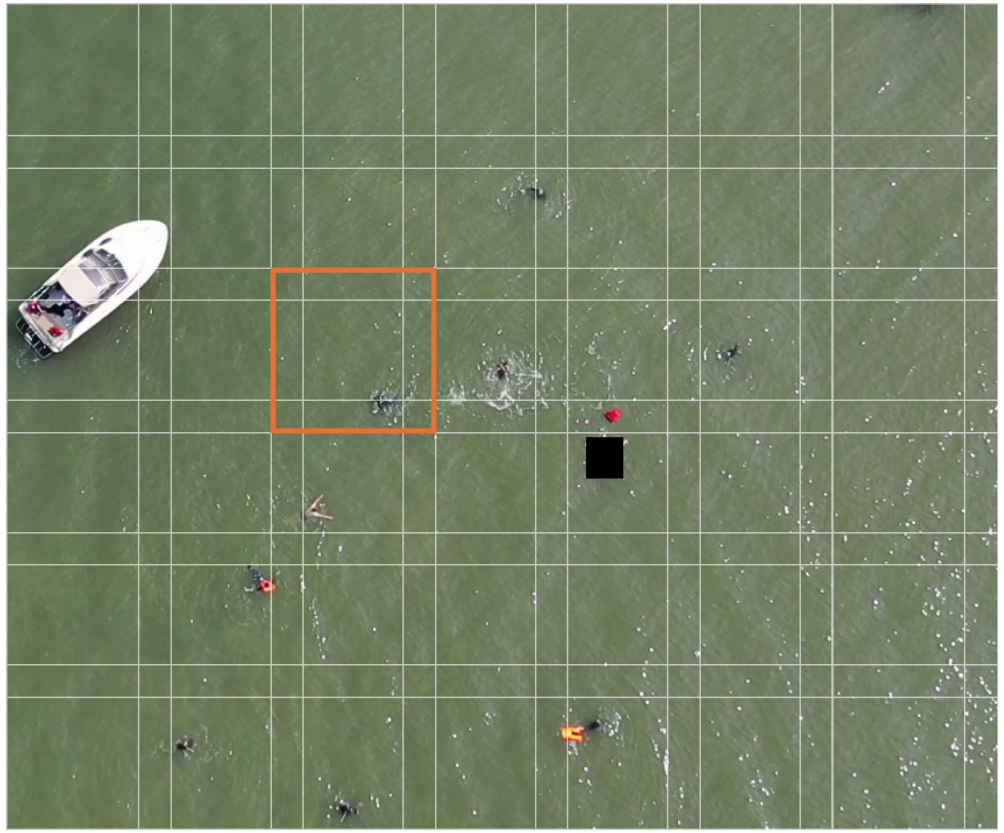


0.60 × (1 − 0.65) = 0.21

Hard-NMS: removes B if IoU ≥ threshold. **Linear Soft-NMS:** $s \leftarrow s \cdot (1 - \text{IoU})$ when IoU ≥ 0.5. Illustrative example.

Hard-NMS is all or nothing; Soft-NMS lowers the confidence with the overlap

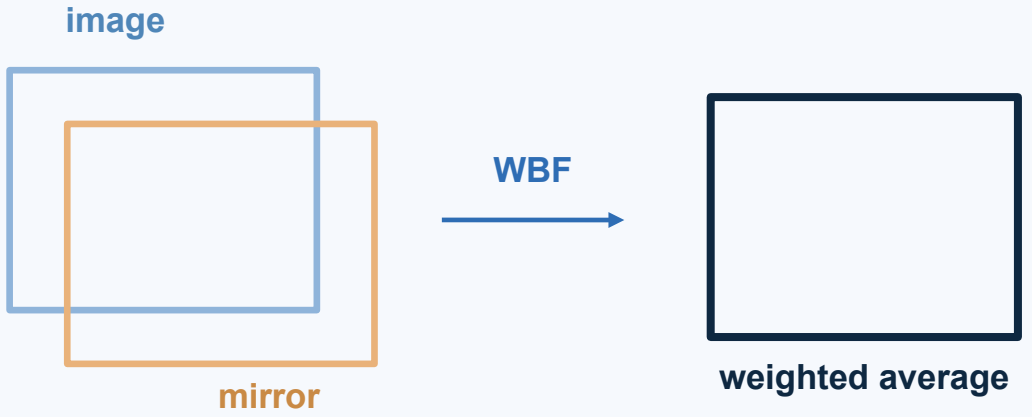
SAHI AND FLIP + WBF



SAHI: 640-px tiles at native resolution, overlap 0.20

Illustrative grid.

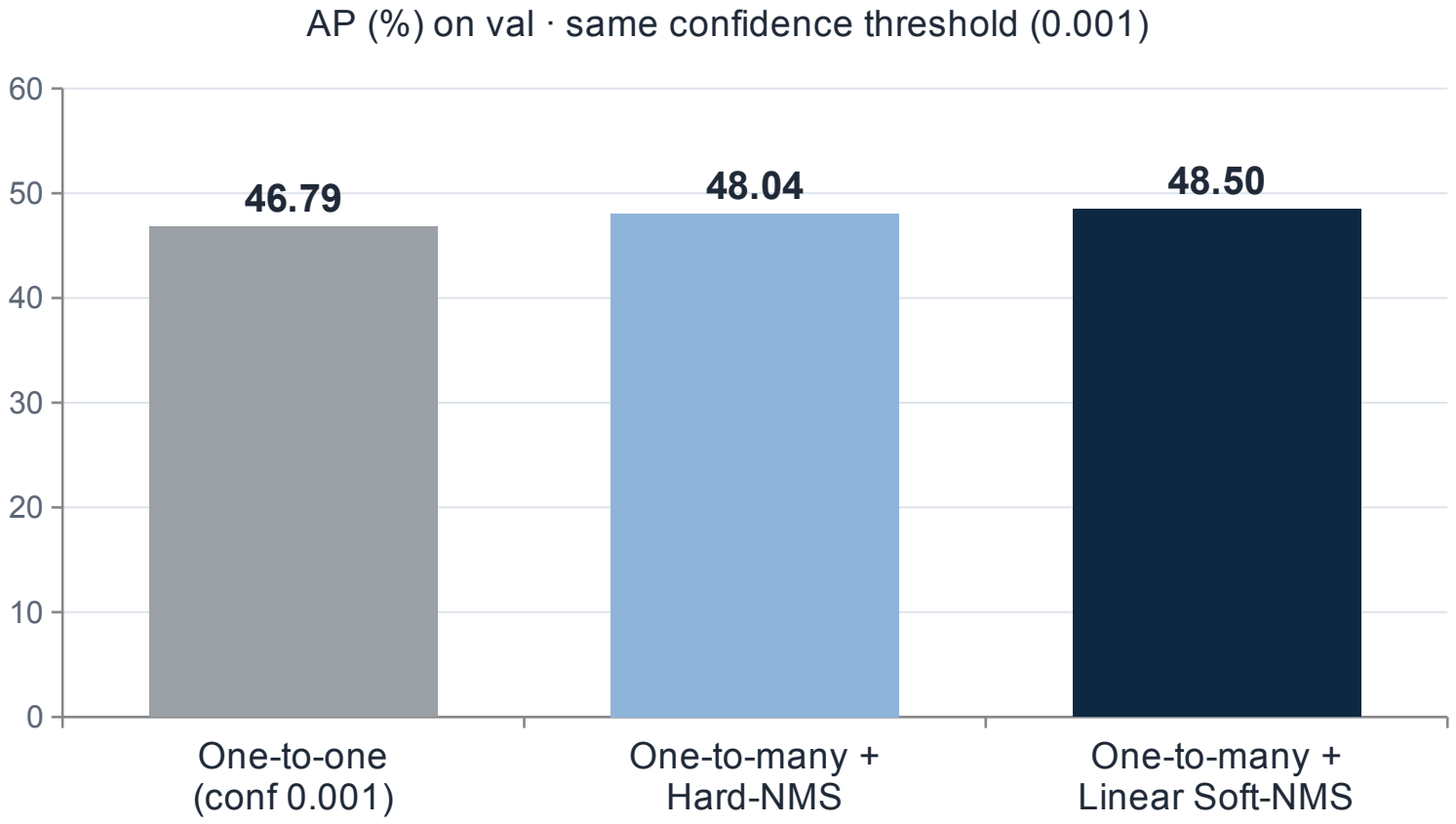
Flip + WBF



WBF averages overlapping boxes instead of discarding them.

More forward passes per image: 2 for Flip + WBF, about 39 for SAHI

RESULT: HEAD AND NMS AT 1280 PX



+1.25 AP

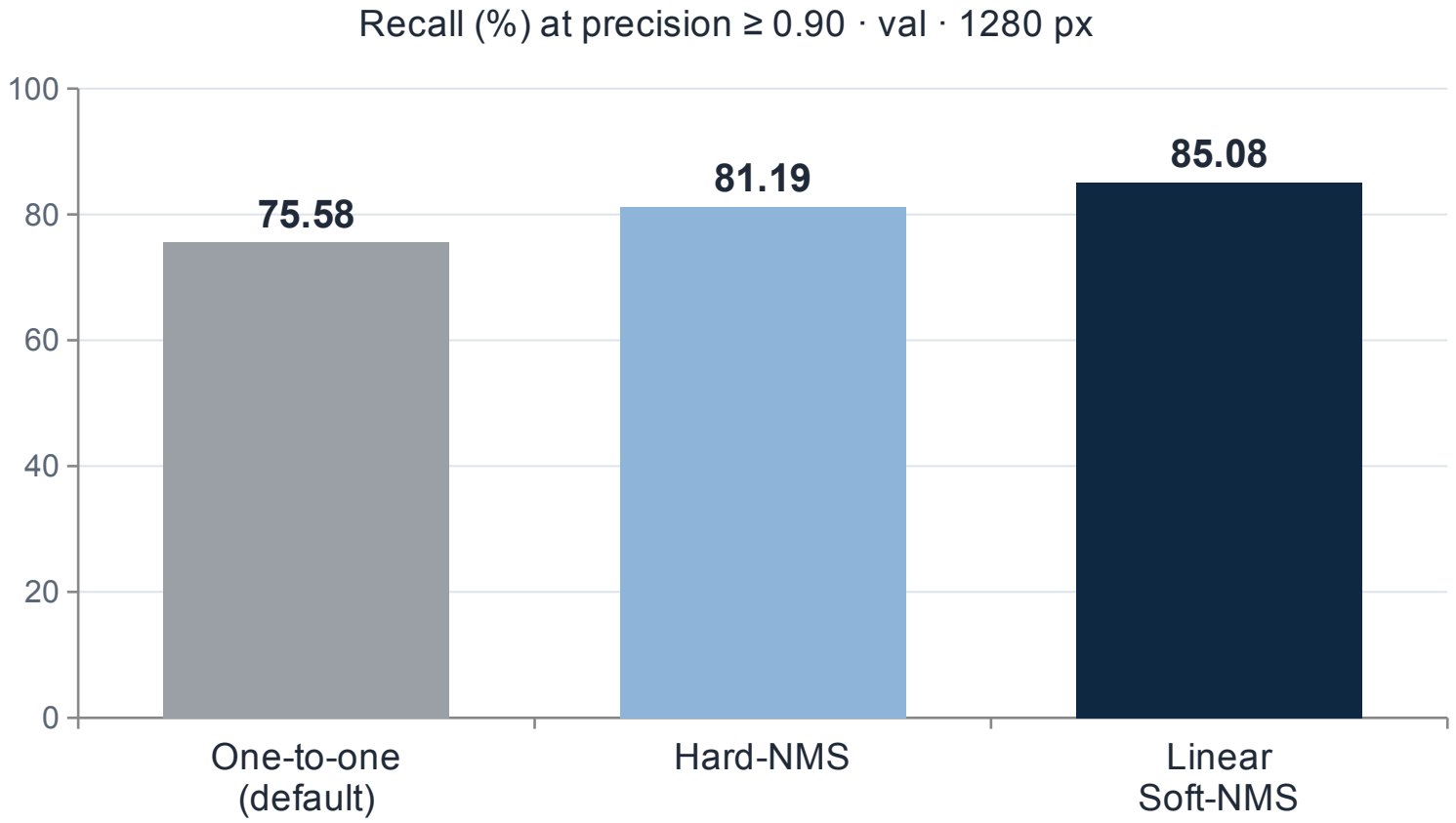
one-to-many head + Hard-NMS vs. one-to-one

+0.46 AP

Soft-NMS vs. Hard-NMS
95% CI [+0.37, +0.55]

The head with NMS beats the default; Soft-NMS adds a small but consistent gain

AT THE OPERATING POINT



87.80%

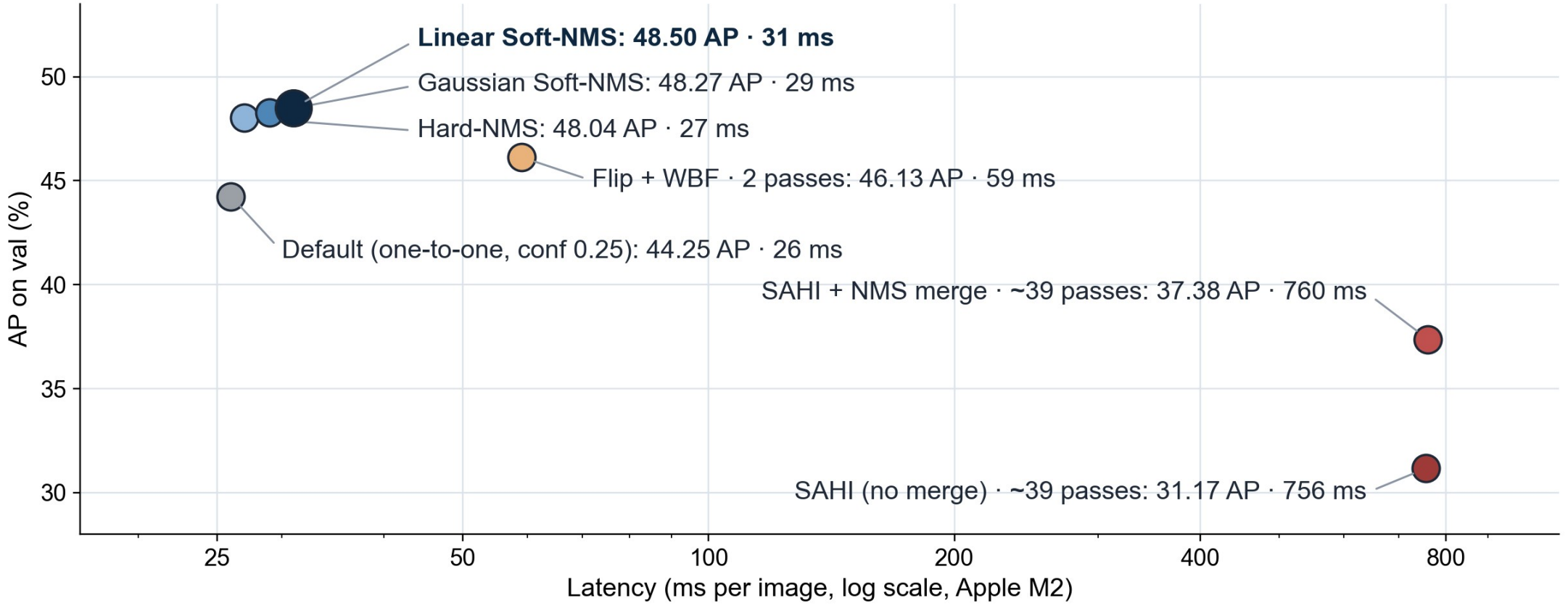
best F1 (Soft-NMS); 86.23% Hard-NMS and 84.73% one-to-one

77.48 → 87.57%

precision at conf 0.25: Hard-NMS → Soft-NMS

At precision ≥ 0.90, Soft-NMS recovers 85.08% of the objects

COST VS. ACCURACY



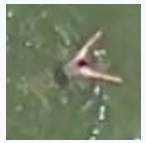
SAHI and WBF: more passes, more latency and lower AP

WHY DOES SAHI LOSE?

HYPOTHESIS

Training and single pass

3840 × 2160 frame → 1280 px



target as the detector learned it

≈ 3×

SAHI tile

640 × 640 crop at native resolution



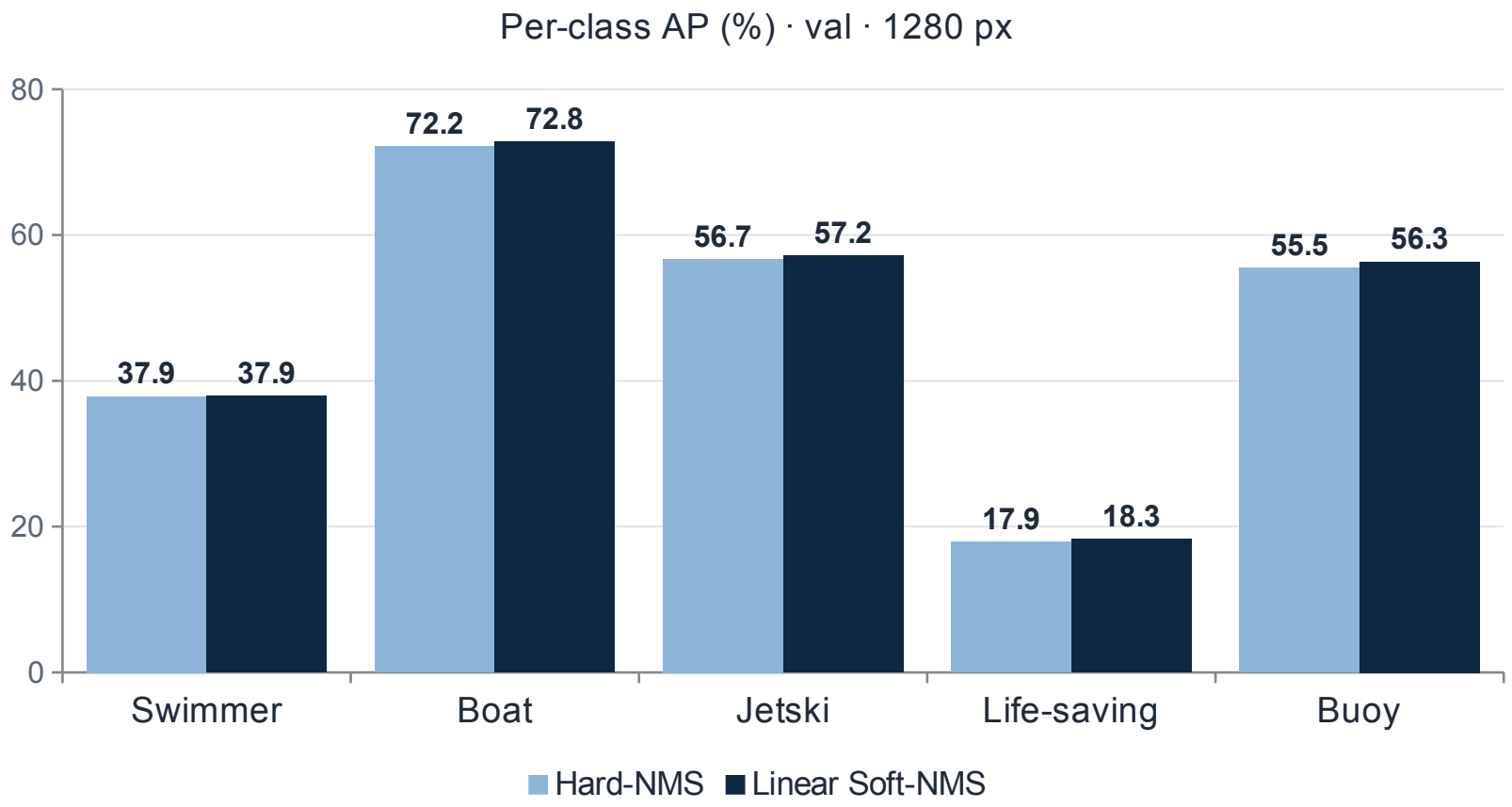
the same target, ~3× larger

Consistent with: largest losses for large objects (API) and boats.

Not tested: SAHI without slicing-aided fine-tuning.

Hypothesis: in the tiles, objects appear about 3× larger than in training

PER-CLASS RESULT



18.26%

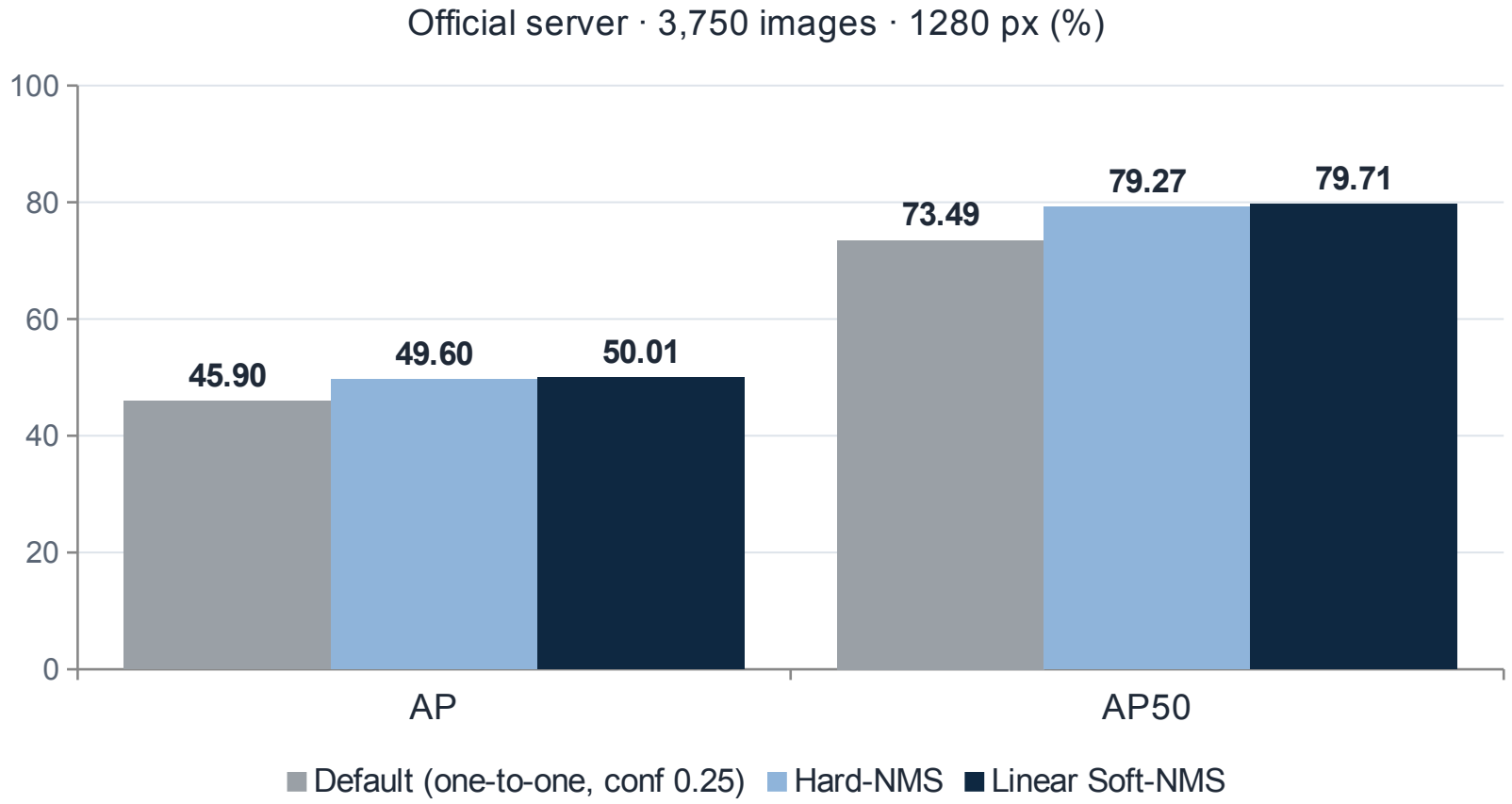
AP of life-saving appliances (Soft-NMS)

5 of 5

classes improve with Soft-NMS

Life-saving appliances remain the hardest class in every configuration

OFFICIAL SEADRONESSEE TEST SET



+4.11 AP

Linear Soft-NMS vs. Default

52.53

AP75 of the final configuration

On the official test set: 45.90 → 50.01 AP and 73.49 → 79.71 AP50

WHERE DOES THE GAIN COME FROM?

Validation: 44.25 → 48.50 AP (+4.25 points)



confidence threshold
0.25 → 0.001

protocol effect

one-to-many head
+ Hard-NMS

Linear
Soft-NMS

Above conf 0.25 the detections are identical: AP integrates the whole precision–recall curve. On the test set, Soft-NMS adds +0.41, within the validation CI.

More than half of the gain is a protocol effect, not Soft-NMS

LIMITATIONS

Constraint	Consequence
One training seed per checkpoint	Training variance not estimated
Latency on a non-dedicated laptop (Apple M2)	No measurement on embedded hardware
SAHI without slicing-aided fine-tuning	The negative result holds for this version
IoU 0.80 tuned on the 640-px lineage	Reused at 1280 px without re-tuning
Operating points estimated on val	No separate split to choose the threshold

Findings specific to YOLO26n on SeaDronesSee, under the tested protocol

CONCLUSIONS

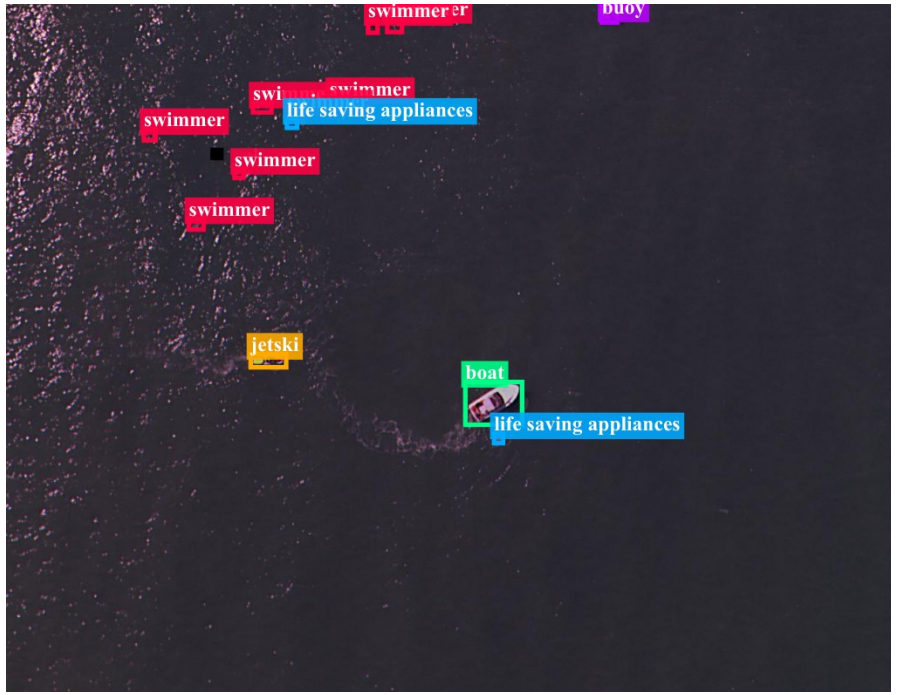
1. **A single pass at 1280 px with Linear Soft-NMS** is the best trade-off: 50.01 AP on the official test set

2. **SAHI and WBF** increase cost without improving AP

3. **The protocol matters:** 2.54 of the 4.25 points come from the confidence threshold alone

Still open: life-saving appliances, at 18.26% AP

THANK YOU



Thank you!

Questions?

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Code and parameters
github.com/IghorRibeiro/
yolo26n-seadronessee-postprocessing

BACKUP: CONFIGURATIONS AT 1280 PX (VAL)

Config	Passes	ms/img	AP	AP50	AP75	APs	APm	APl
Default (one-to-one, conf 0.25)	1	26	44.25	74.98	43.67	40.59	45.69	61.28
Hard-NMS	1	27	48.04	79.98	48.23	43.13	48.28	66.41
Linear Soft-NMS	**1**	**31**	**48.50**	**80.36**	**48.66**	**43.65**	**48.53**	**66.87**
Gaussian Soft-NMS	1	29	48.27	79.97	48.41	43.57	48.30	66.68
Flip + WBF	2	59	46.13	79.45	44.15	40.16	47.15	65.18
SAHI (no merge)	38.63	756	31.17	51.90	32.58	36.84	34.69	42.36
SAHI + NMS merge	38.63	760	37.38	64.66	37.76	41.33	43.41	45.70

Latency: Apple M2, 8 GB, MPS, batch 1, 200 validation images, after 10 warm-up iterations, image decoding excluded.

Single pass: 26 to 31 ms per image; cost is set by the number of forward passes

BACKUP: HISTORICAL 640-PX CHECKPOINT (VAL)

Config	Input (px)	AP	AP50	AP75
One-to-one	640	39.09	70.55	37.19
One-to-many + Hard-NMS	640	39.78	71.09	38.22
Native TTA	640	37.73	68.30	36.15
MS-Flip + WBF	640, 800	42.51	75.35	39.83
SAHI-640	640; tiles 640	37.93	66.82	37.49
SAHI-1024	640; tiles 1024	37.54	65.71	37.50
Full image + SAHI-640 + WBF	640; tiles 640	37.63	66.52	36.45
Hard-NMS	800	42.83	74.27	42.49
Linear Soft-NMS	**800**	**43.25**	**75.30**	**42.85**

At 640 px too, TTA and SAHI do not beat full-image inference with NMS

BACKUP: PER-CLASS AP AT 1280 PX (VAL)

Config	Swimmer	Boat	Jetski	Life-saving	Buoy
Hard-NMS	37.86	72.20	56.71	17.92	55.50
Linear Soft-NMS	**37.88**	**72.82**	**57.21**	**18.26**	**56.32**
Gaussian Soft-NMS	37.56	72.51	57.09	18.24	55.95
Flip + WBF	32.52	72.75	55.04	17.59	52.72
SAHI (no merge)	28.14	43.34	31.51	14.87	38.00
SAHI + NMS merge	33.11	50.97	39.38	17.70	45.73

With SAHI, the largest loss is on boats, the class with the largest boxes